Unit 1, Lesson 9: Adding and Subtracting Radical Expressions

Benchmark

MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.

Learning Target

☐ I can add or subtract two square roots or two cube roots.

1.9.1 Warm-Up: Adding and Subtracting Algebraic Expressions

1. Find the sum or difference of each algebraic expression.

$4x + 23x$ $\quad$ $-7t - 40t$ $\quad$ $-35b + 10b$

2. Complete the statement to explain why is it possible to combine the terms in each of these expressions.

When adding and subtracting algebraic expressions, __________ terms can be combined.

3. Which, if any, of the radical expressions do you think can be combined?

☐ $\sqrt{17} + \sqrt{32}$
☐ $5\sqrt{13} - 8\sqrt{3}$
☐ $-7\sqrt{17} + 3\sqrt{17}$
☐ $9\sqrt{5} + \sqrt{5}$
☐ $8\sqrt{40} - 4\sqrt{9}$
☐ $11\sqrt{48} - 6\sqrt{48}$
☐ $-2\sqrt{49} - 3\sqrt{4}$

1.9.2 Exploration: Adding and Subtracting Radical Expressions

1. In small groups, determine the value of each expression.

   $\sqrt{25} + \sqrt{100}$ $\qquad\qquad$ $\sqrt{25+100}$

2. Explain which of the expressions is equivalent to $\sqrt{125}$.

3. Paloma says $\sqrt{16} - \sqrt{9} = \sqrt{7}$. Ty'rique says $\sqrt{16} - \sqrt{9} = 1$. Who is correct? Verify by entering $\sqrt{16} - \sqrt{9}$ into a calculator.

   ____________ is correct because $\sqrt{16} - \sqrt{9}$ is equal to

   _____.

4. Xavier says $\sqrt{12} + 2\sqrt{12} = 3\sqrt{12}$. Sam says $\sqrt{12} + 2\sqrt{12} = 2\sqrt{24}$. Who is correct? Verify using a calculator.

   ____________ is correct because $\sqrt{12} + 2\sqrt{12}$ is equal to

   _______.

1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions

When adding and subtracting algebraic expressions, only ***like terms*** can be combined. Adding and subtracting radical expressions is similar because only like radicals can be combined.

Like radicals are radical terms that have the same index and the same radicand.

1. Find the sum or difference of each radical expression.

   a. $4\sqrt{5} + 7\sqrt{5} =$ ____________

   b. $30\sqrt[3]{2} - 23\sqrt[3]{2} =$ ____________

   c. $-10\sqrt{7} - 23\sqrt[3]{7} =$ ____________

   d. $-\sqrt{17} + \sqrt{17} =$ ____________

Three steps can be used when adding and subtracting radical expressions.

Step 1: Find perfect squares (for square roots) or perfect cubes (for cube roots) that are factors of the radicand.

Step 2: Rewrite the radical expressions by applying the product law of radicals and evaluating any perfect squares or cubes.

Step 3: Add or subtract like radical terms.

2. Use the steps to add the following radical expression.

$$3\sqrt{6} + 5\sqrt{24}$$

Step 1: Both terms include square roots. If possible, find perfect squares that are factors of the radicands, 6 and 24.

Step 2: Rewrite the radical expressions by applying the product law of radicals and evaluating any perfect squares.

$$3\sqrt{6} + 5\sqrt{24} = \underline{\qquad\qquad} + \underline{\qquad\qquad}$$

Step 3: Add or subtract like radical terms.

3. Three students added $\sqrt{180}$ and $\sqrt{45}$ differently. Their work is shown.

Gerald's Work	Kalo's Work	Paola's Work
$\sqrt{180} + \sqrt{45}$	$\sqrt{180} + \sqrt{45}$	$\sqrt{180} + \sqrt{45}$
$2\sqrt{45} + \sqrt{45}$	$3\sqrt{20} + 3\sqrt{5}$	$6\sqrt{5} + 3\sqrt{5}$
$3\sqrt{45}$		$9\sqrt{5}$

a. Use a calculator to verify if the students' answers are equivalent.

b. Why did the students end up with different expressions for their sum?

c. Show how Gerald and Kalo could continue their work to arrive at the same expression as Paola.

Gerald's Next Steps	Kalo's Next Steps
$3\sqrt{45}$	$3\sqrt{20} + 3\sqrt{5}$

4. Megh showed her work on the problem.

$$\sqrt[3]{32} + \sqrt[3]{64} = \sqrt[3]{8 \cdot 4} + 4 = 2\sqrt[3]{4} + 4 = 6\sqrt[3]{4}$$

Write an explanation to Megh explaining the error in her work.

1.9.4 Guided Practice: Adding and Subtracting Radical Expressions

1. Select all of the expressions that can be combined. For the selected expressions, combine the terms. Show your work.

- ☐ $\sqrt{17} + \sqrt{32}$
- ☐ $5\sqrt{13} - 8\sqrt{3}$
- ☐ $-7\sqrt{17} + 3\sqrt{17}$
- ☐ $9\sqrt{5} + \sqrt{5}$
- ☐ $8\sqrt{40} - 4\sqrt{9}$
- ☐ $11\sqrt{48} - 6\sqrt{48}$
- ☐ $-2\sqrt{49} - 3\sqrt{4}$

2. With a partner, find the sum and difference of the following problems. If you cannot add or subtract, circle the expression.

a. $\sqrt[3]{6} + 3\sqrt{6}$

b. $\sqrt{20} - 3\sqrt{45}$

c. $16^{\frac{1}{3}} - 2^{\frac{1}{3}}$

d. $\sqrt{72} + \sqrt{18} - \sqrt{15}$

1.9.5 Your Turn!

1. Find the sum or difference. If you cannot add or subtract, circle the expression.

a. $\sqrt[3]{27000} - \sqrt[3]{512}$

b. $\sqrt{50} + \sqrt{18} + \sqrt{10}$

c. $\sqrt[3]{6} - \sqrt[3]{48}$

d. $32^{\frac{1}{2}} - 16^{\frac{1}{3}}$

1.9.6 Wrap-Up: Ticket Out the Door

1. Find the sum.

$$17\sqrt{8} + 5\sqrt{8}$$